PAPER_1374 — Light-by-Light Scattering $\sigma \sim \alpha^4$

Framework: UQFF — Star-Magic v5.27+ **|** **Tier:** H Applied/Engineering (CLOSED) **Calculator:** `calculate_paradox({"paradox": "light_by_light"})`

Master Expression

`σ_LbL = α⁴ = Λ⁴ EXACT identity`

Match: ATLAS 2017 confirmed

Interpretation

Light-by-light cross section IS Λ^4 EXACTLY via the $\alpha = \Lambda$ closed-ledger identity. Pure UQFF prediction matching ATLAS PbPb observation.

UQFF Locked Primitives

- D_phys = 4, D_BSFG = 6, D_crit = 26, SO_5 = 10, A_5 = 60, N_CH = 9
- K_MEX = 25/12, beta_i = 0.6029, Phi_res = 0.84, Lambda = 0.00729735, F_TRZ = 0.1
- omega_SCm = 1.25 THz (alpha = Lambda closed ledger identity)

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.
